

Align your Latents: High-Resolution Video Synthesis with Latent Diffusion Models

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Overview

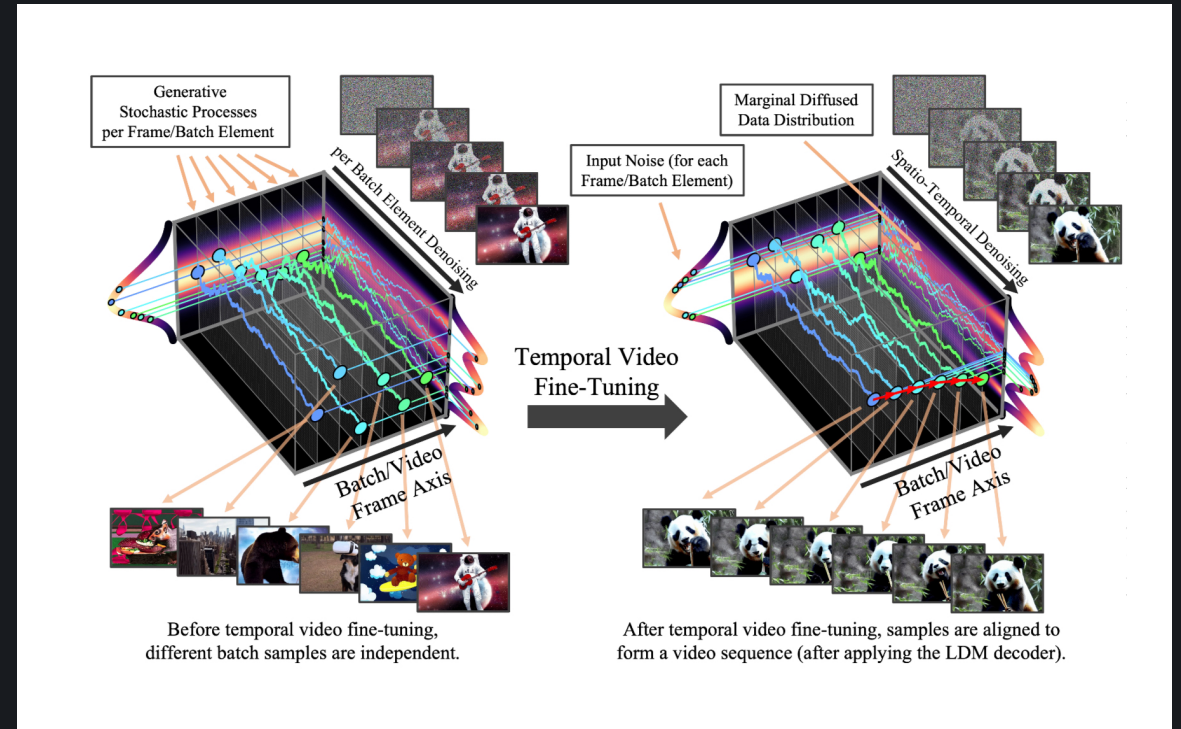
Paper info

CVPR 2023

Key words: **Video Generation, LDM**

Authors: **Blattmann et al.**

Institution: **LMU Munich, NVIDIA**



The key issue that the above paper focuses on

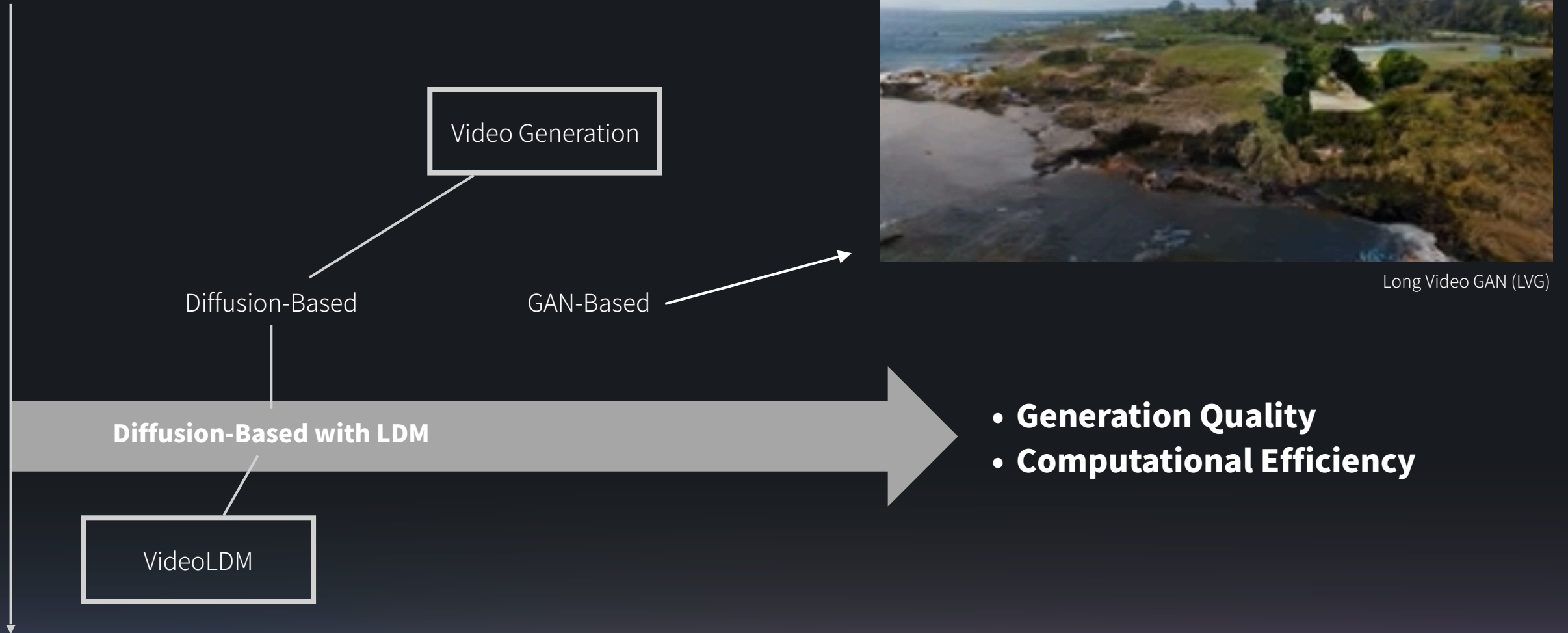
The lack of video training sources →

Reuses and fine-tunes pre-trained image generation model adding additional temporal layer

High computational cost →

Operates based on latent space with CFG and temporal interpolation for high frame rates

Positioning



Long Video GAN (LVG)

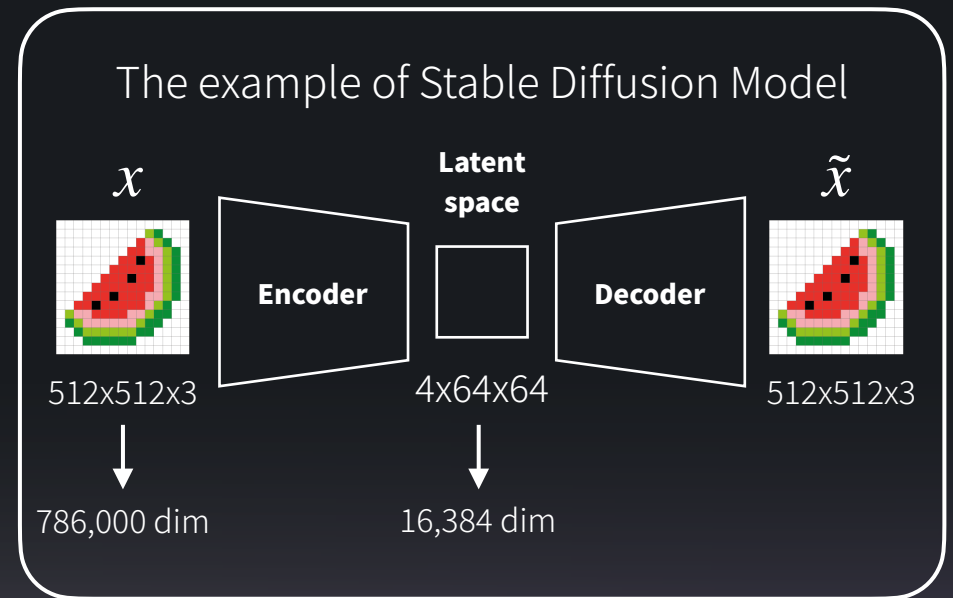
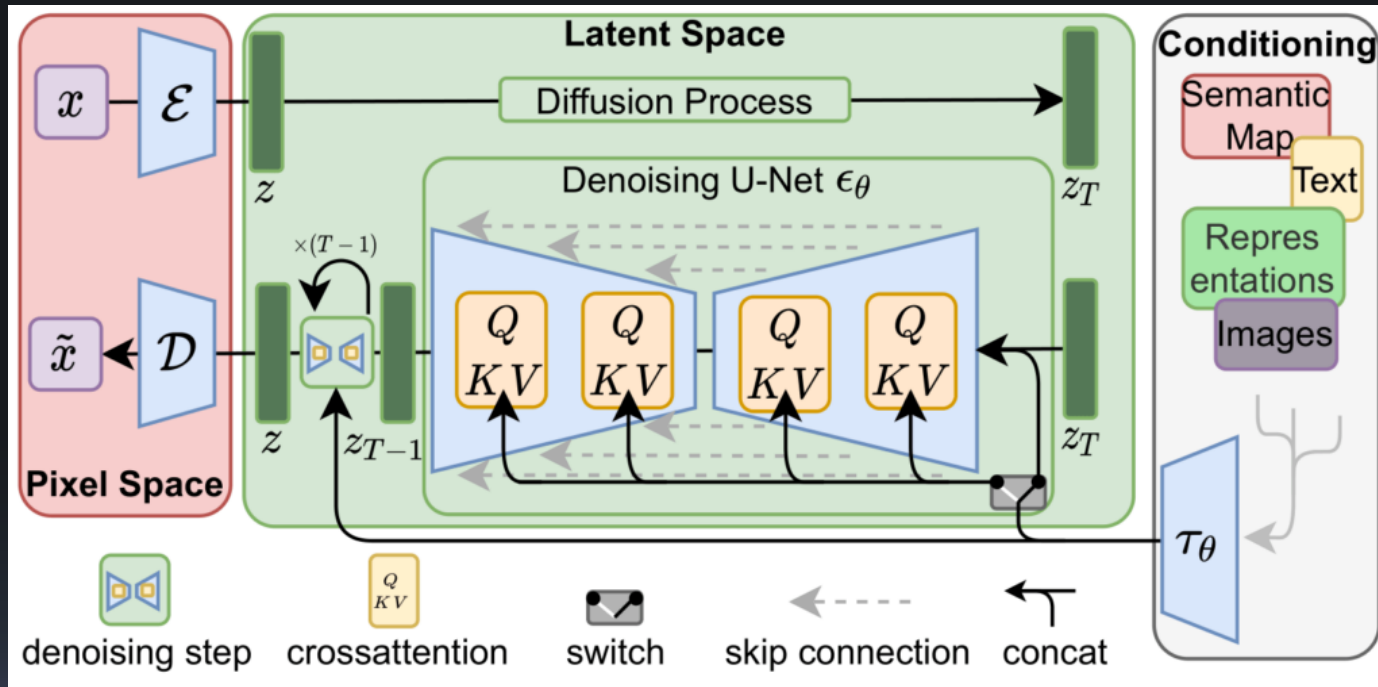
Preliminary Latent Diffusion Model (LDM)

$$L_{\text{DM}} = \mathbb{E}_{x, \epsilon \sim \mathcal{N}(0,1), t} \left[\left\| \epsilon - \epsilon\theta(x_t, t) \right\|_2^2 \right]$$

Directly adding noise to pixel space

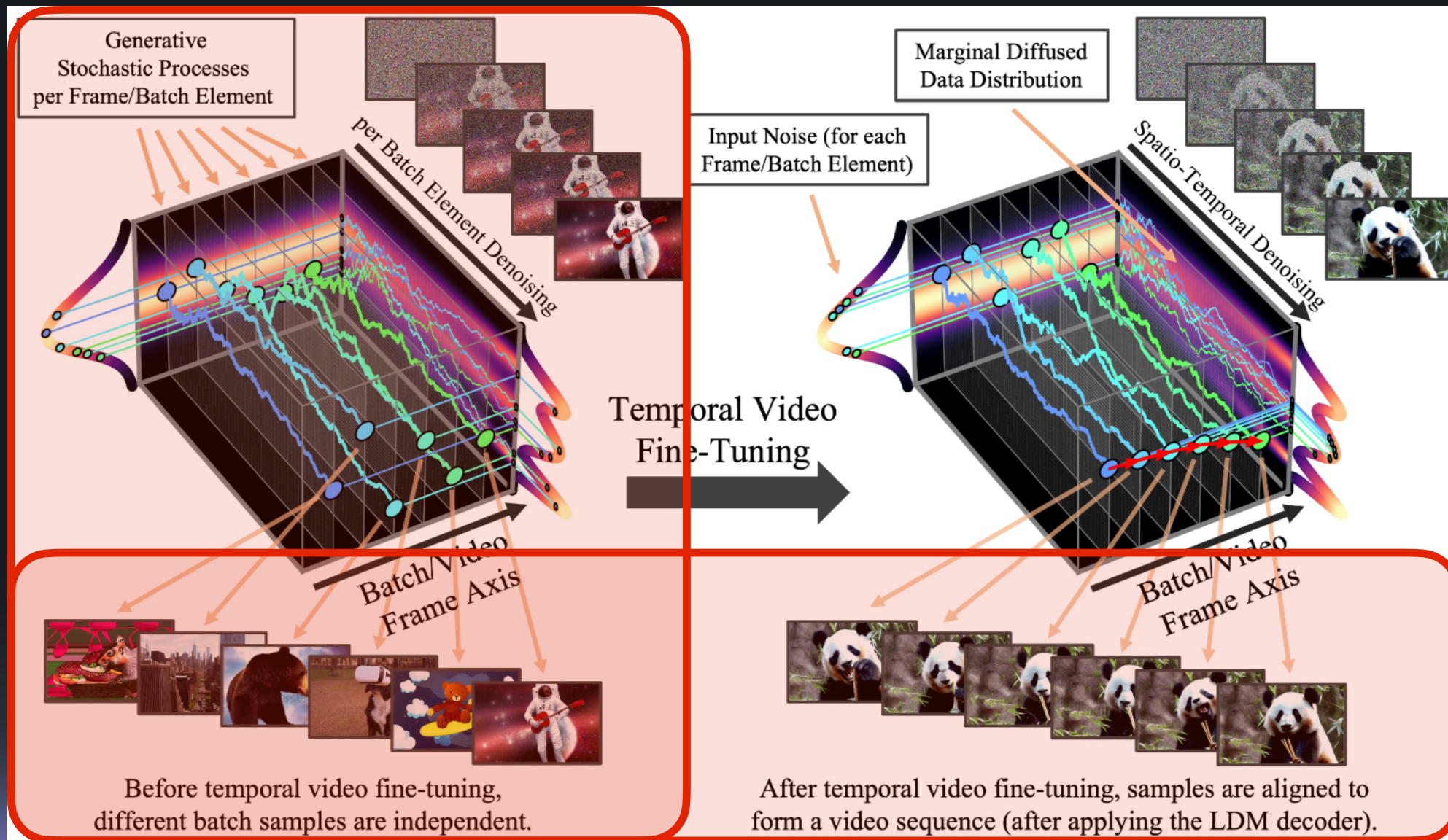
$$L_{\text{LDM}} := \mathbb{E}_{\mathcal{E}(x), \epsilon \sim \mathcal{N}(0,1), t} \left[\left\| \epsilon - \epsilon\theta(z_t, t) \right\|_2^2 \right]$$

Adding noise to encoded latent space

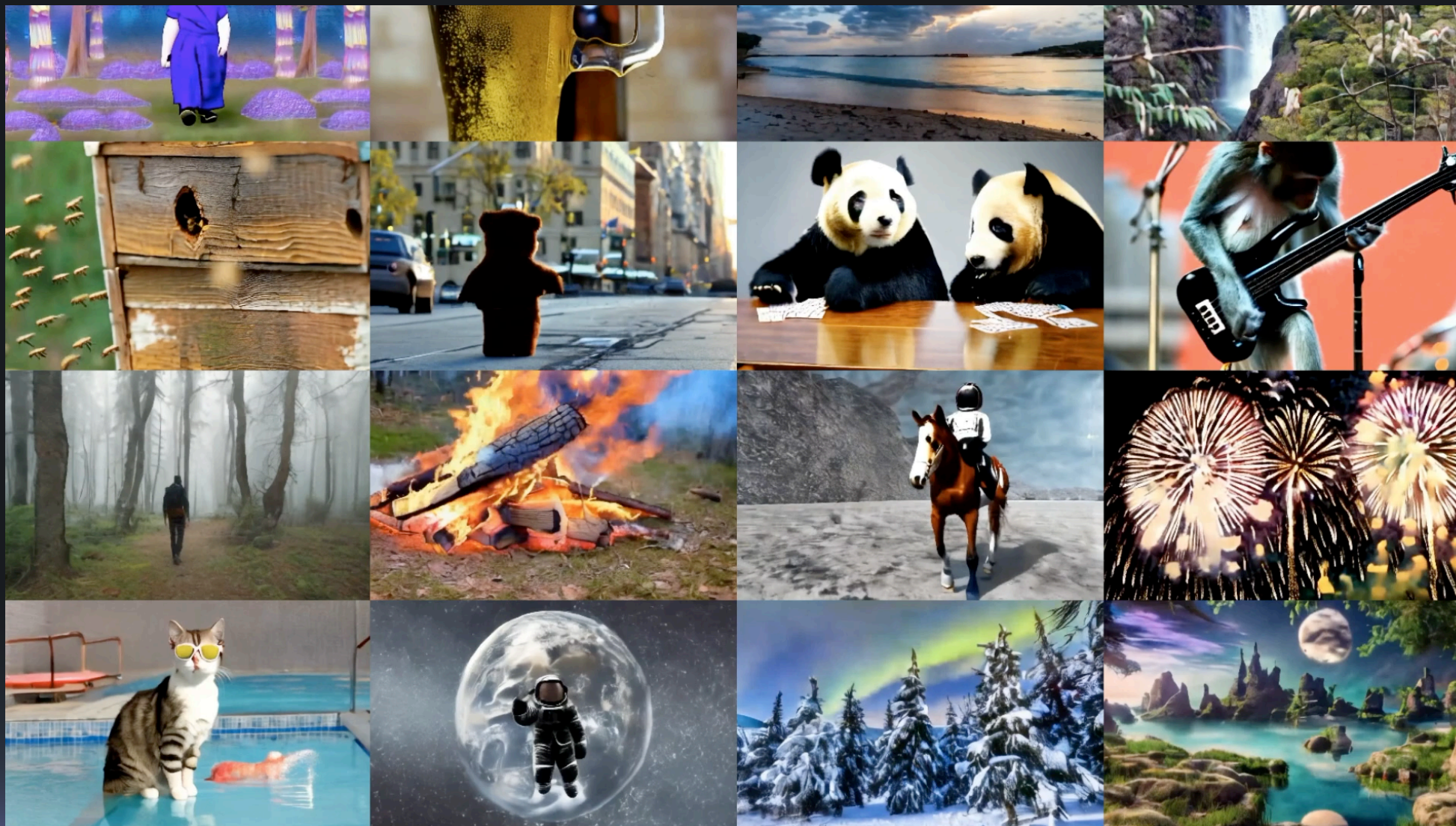


Overview

Latent
Diffusion
Model
(Stable
Diffusion
Model)



Results



1280 x 2048 pixels, 113 frames, 24 fps 4.7 sec

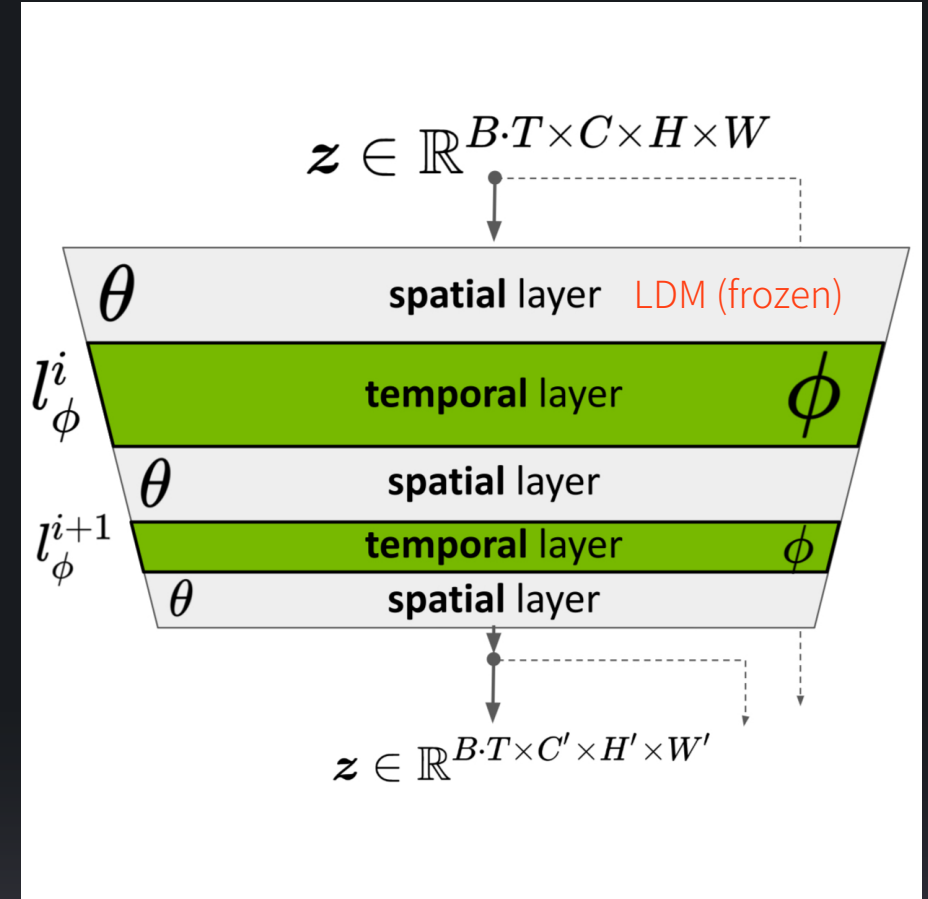
Method **Objective Term**

$$\arg \min_{\phi} \mathbb{E}_{x \sim p_{\text{data}}, \tau \sim p_T, \epsilon \sim \mathcal{N}(0, I)} \left[\left\| \mathbf{y} - \mathbf{f}_{\theta, \phi}(\mathbf{z}_{\tau}; \mathbf{c}, \tau) \right\|_2^2 \right]$$

ϕ
 learnable parameter

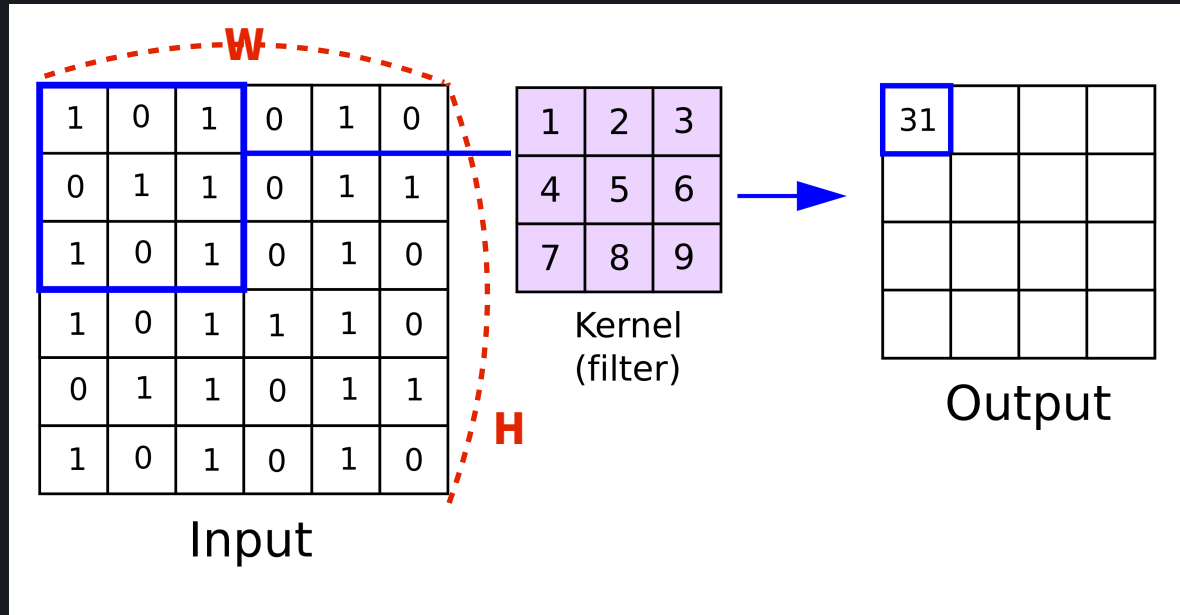
Noise prediction target
 $\mathbf{y}$

Condition (prompt)
 $\mathbf{c}$

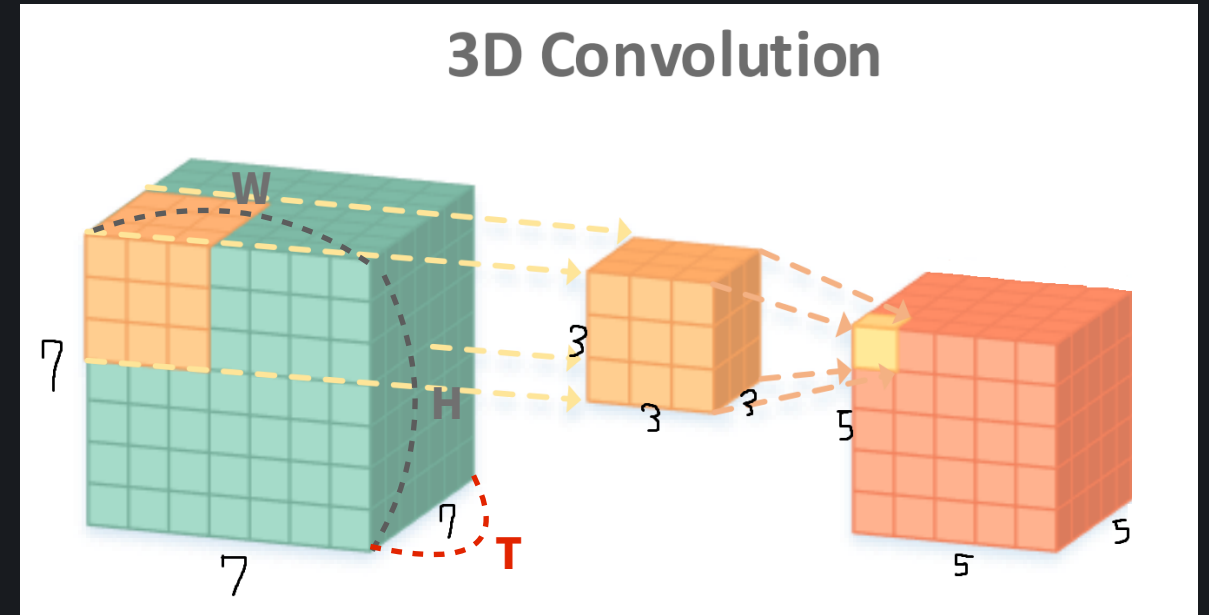


i: layer index

Preliminary 3D Convolution

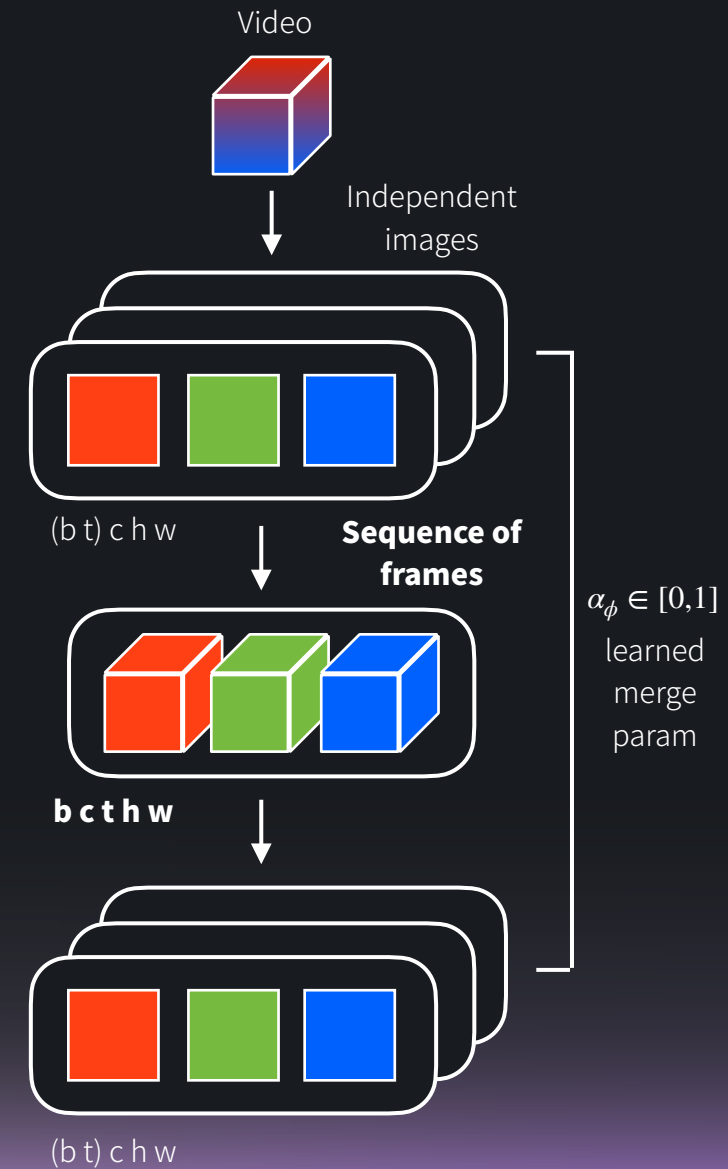
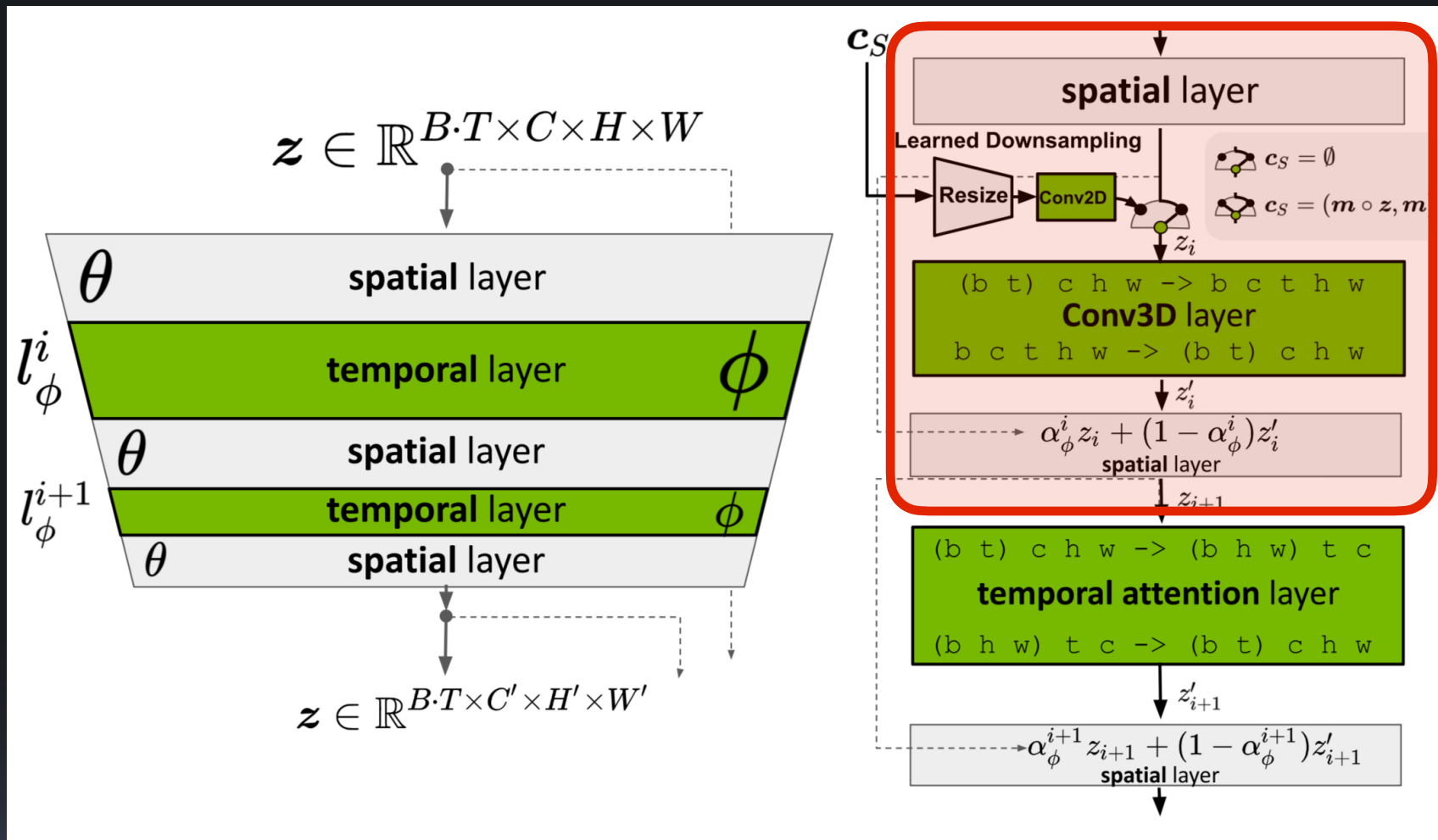


2D Convolution
(Image)

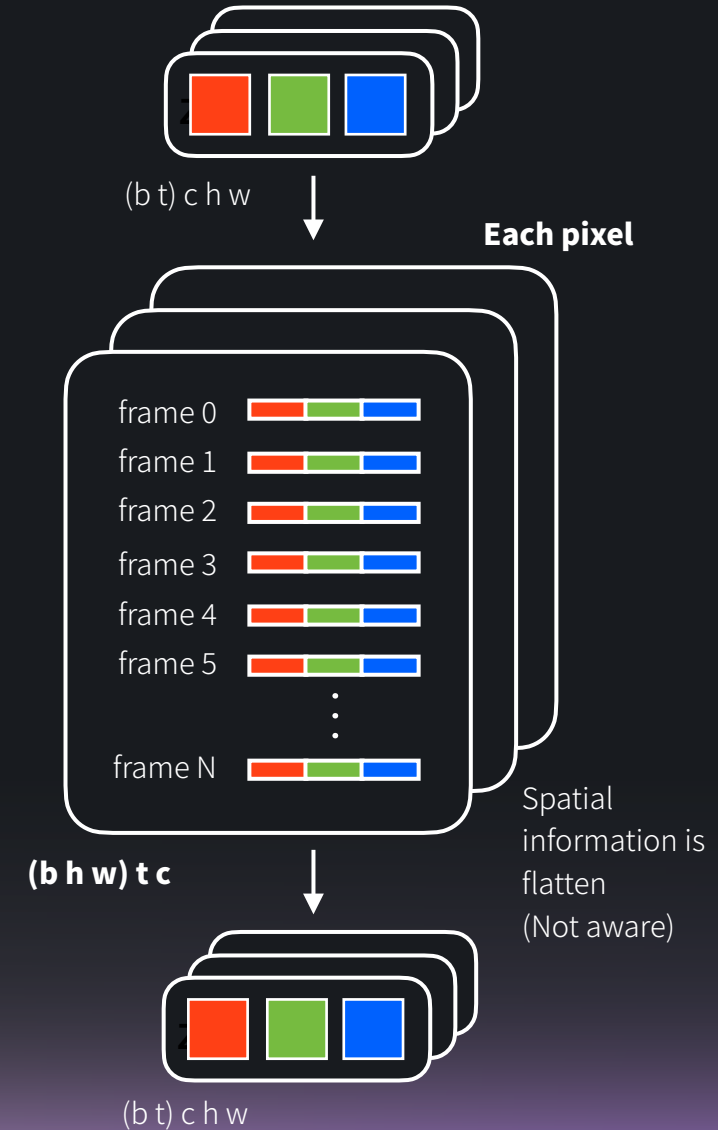
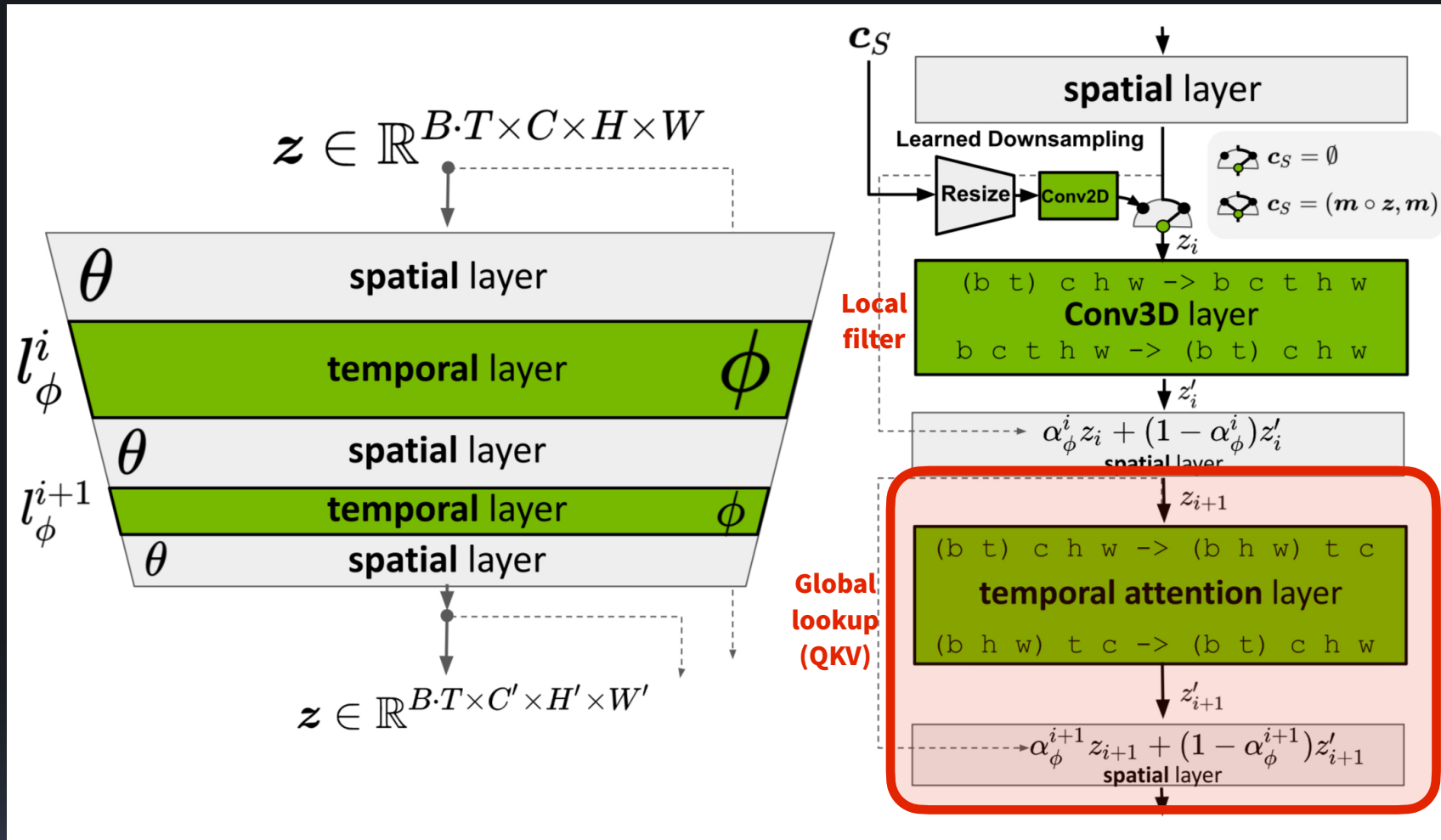


3D Convolution
(Video)

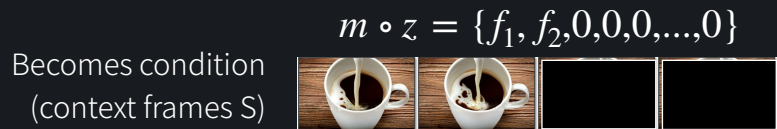
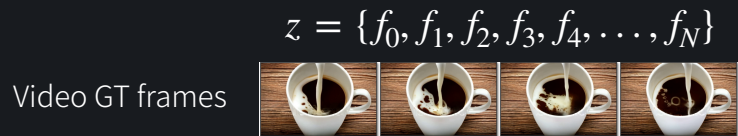
Method Training Phase



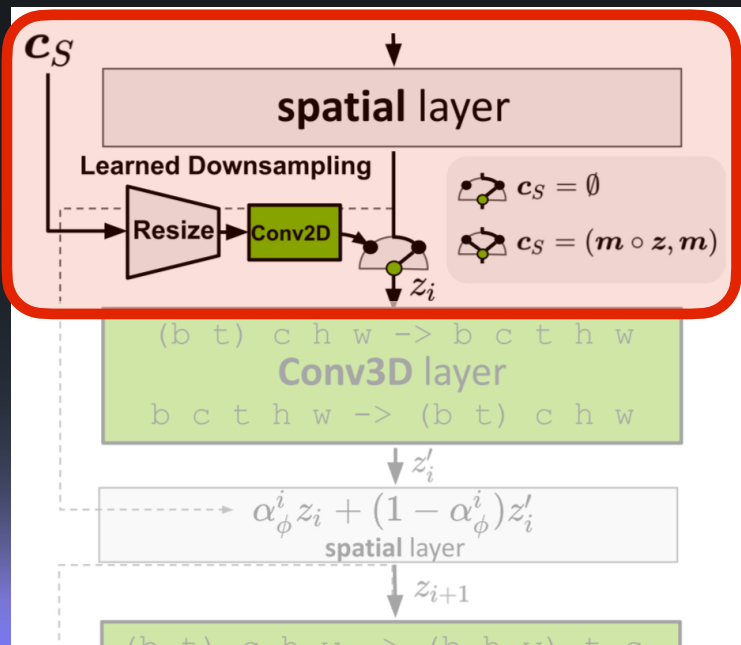
Method Training Phase



Method Training Phase



Fill-in-the-blank Task



For long-term video generation

Noise prediction **without** context frame condition

$$\mathbb{E}_{x \sim p_{\text{data}}, \tau \sim p_T, \epsilon \sim \mathcal{N}(0, I)} \left[\left\| \mathbf{y} - \mathbf{f}_{\theta, \phi}(\mathbf{z}_\tau; \mathbf{c}, \tau) \right\|_2^2 \right]$$

Noise prediction **with** context frame condition

$$\mathbb{E}_{x \sim p_{\text{data}}, \mathbf{m}_S \sim p_S, \tau \sim p_T, \epsilon \sim \mathcal{N}(0, I)} \left[\left\| \mathbf{y} - \mathbf{f}_{\theta, \phi}(\mathbf{z}_\tau; \mathbf{c}_S, \mathbf{c}, \tau) \right\|_2^2 \right]$$

Guidance scale

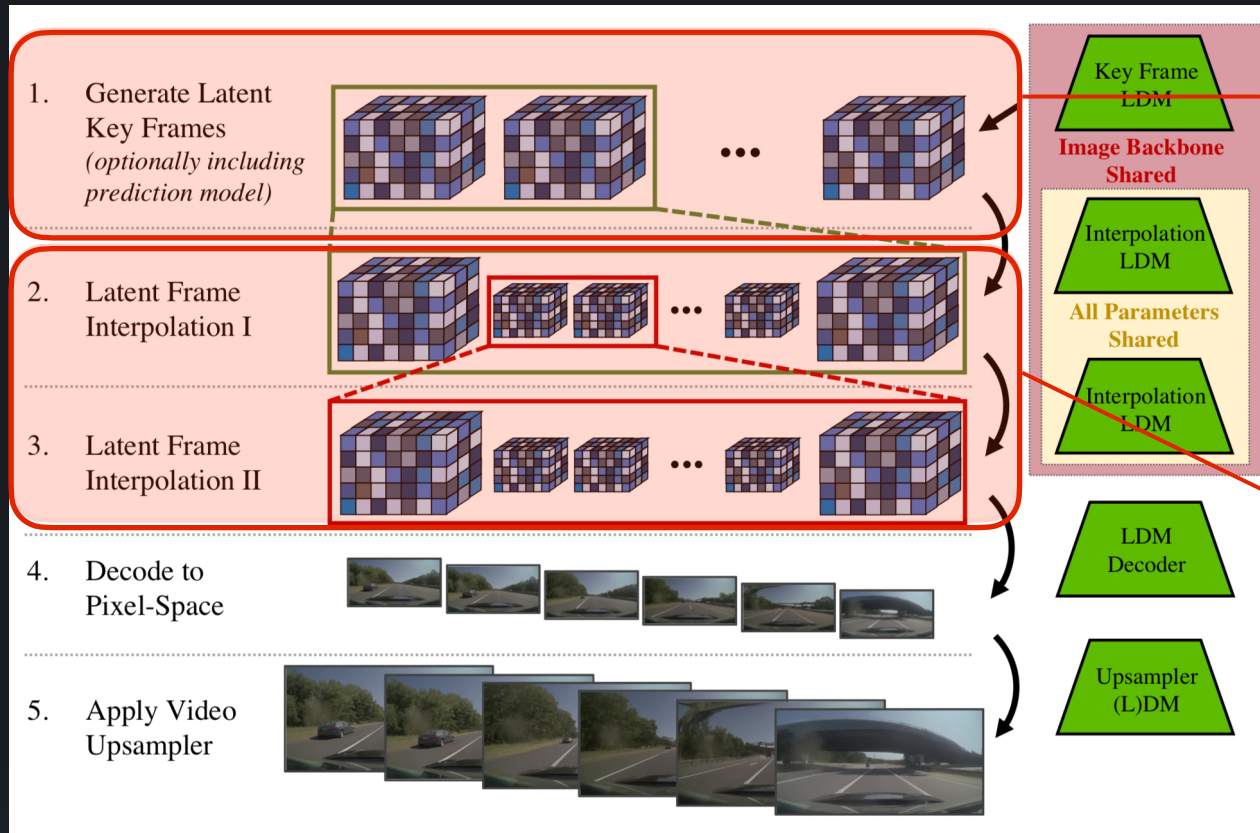
$$\mathbf{f}'_{\theta, \phi}(\mathbf{z}_\tau; \mathbf{c}_S) = \mathbf{f}_{\theta, \phi}(\mathbf{z}_\tau) + s \cdot (\mathbf{f}_{\theta, \phi}(\mathbf{z}_\tau; \mathbf{c}_S) - \mathbf{f}_{\theta, \phi}(\mathbf{z}_\tau))$$

Classifier-free diffusion guidance (CFG)

Method Inference Phase

Low
Frame
Rate

High
Frame
Rate

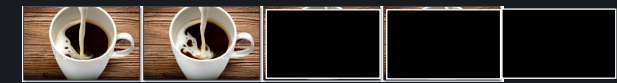


Predict the future
(Generate key frames)

$$m = \{1, 1, 0, 0, 0, \dots, 0\}$$

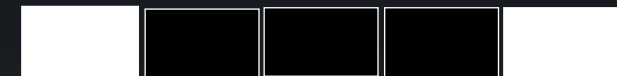


$$m \circ z = \{f_1, f_2, 0, 0, 0, \dots, 0\}$$

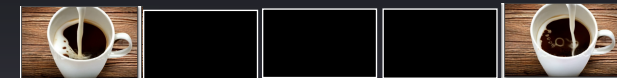


Interpolate in-between frames

$$m = \{1, 0, 0, 0, 1\}$$



$$m \circ z = \{f_0, 0, 0, 0, f_4\}$$



Validation

Dataset for fine-tuning

- Real Driving Scene (RDS)
- WebVid-10M
- Mountain Biking dataset
- UCF-101 and MSR-VTT

Metrics

- **Fréchet Inception Distance (FID)**: Used to evaluate the quality of individual video frames (frame-wise).
- **Fréchet Video Distance (FVD)**: Used to measure the quality and temporal consistency of the generated videos.
- **Human Evaluation**: Human participants were asked to choose between video samples.
- **CLIP Similarity (CLIP-SIM)**: Used for the text-to-video experiments to measure how well the generated video matches the text prompt.
- **Inception Score (IS)**: Also used for the text-to-video models.

Table 1. *Left*: Comparison with LVG on RDS; *Right*: Ablations.

Method	FVD	FID	Method	FVD	FID
LVG [6]	478	53.5	Pixel-baseline	639.56	59.70
<i>Ours</i>	389	31.6	End-to-end LDM	1155.10	71.26
<i>Ours</i> (cond.)	356	51.9	Attention-only	704.41	50.01
			<i>Ours</i>	534.17	48.26
			<i>Ours</i> (context-guided)	508.82	54.16

Table 2. User study on Driving Video Synthesis on RDS.

Method	Pref. A	Pref. B	Equal
<i>Ours</i> (cond.) v.s <i>Ours</i> (uncond.)	49.33	42.67	8.0
<i>Ours</i> (uncond.) v.s LVG	54.02	40.23	5.74
<i>Ours</i> (cond.) v.s LVG	62.03	31.65	6.33

Table 3. *Left*: Evaluating temporal fine-tuning for diffusion up-samplers on RDS data; *Right*: Video fine-tuning of the first stage decoder network leads to significantly improved consistency.

Method	FVD	FID	Decoder	image-only	finetuned
<i>Ours</i> Image Upsampler	165.98	19.71	FVD	390.88	32.94
<i>Ours</i> Video Upsampler	45.39	19.85	FID	7.61	9.17

Table 4. UCF-101 text-to-video generation.

Method	Zero-Shot	IS (↑)	FVD (↓)
CogVideo (Chinese) [32]	Yes	23.55	751.34
CogVideo (English) [32]	Yes	25.27	701.59
MagicVideo [109]	Yes	-	699.00
Make-A-Video [76]	Yes	33.00	367.23
Video LDM (<i>Ours</i>)	Yes	29.49	656.49

Table 5. MSR-VTT text-to-video generation performance.

Method	Zero-Shot	CLIPSIM (↑)
GODIVA [98]	No	0.2402
NÜWA [99]	No	0.2439
CogVideo (Chinese) [32]	Yes	0.2614
CogVideo (English) [32]	Yes	0.2631
Make-A-Video [76]	Yes	0.3049
Video LDM (<i>Ours</i>)	Yes	0.2848

Discussion

Reasons for Acceptance of This Paper

- Reuses a text-to-image foundation model to address the lack of large-scale video training datasets.
- Showed superior performance in generating videos with:
 - higher resolution,
 - longer duration,
 - and higher frame rates

Limitations of the paper

- Has repetitive pattern
- Lack of geometric plausibility:
 - Unnatural body deformation,
 - and object flickering

